

**DIFFUSION
MASS TRANSFER IN FLUID
SYSTEMS**

THIRD EDITION

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Inserting these values into the above and rearranging, we have

$$\frac{kR}{D} = [\text{constant}] \left(\frac{R^2 \omega}{D} \right)^\beta \left(\frac{d}{R} \right)^\gamma$$

The dimensionless group raised to the β power is a Péclet number; that raised to the γ power is a ratio of lengths.

This correlation will be successful only if we have guessed the right key variables. It may work if the tablets remain intact. It probably won't work if they disintegrate. Still, the analysis gives us a good first step in organizing our results.

8.5 Mass Transfer Across Interfaces

We now turn to mass transfer across interfaces, from one fluid phase to the other. This is a tricky subject, one of the main reasons that mass transfer is felt to be a difficult subject. In the previous sections, we used mass transfer coefficients as an easy way of describing diffusion occurring from an interface into a relatively homogeneous solution. These coefficients involved approximations and sparked the explosion of definitions exemplified by Table 8.2-2. Still, they are an easy way to correlate experimental results or to make estimates using the published relations summarized in Tables 8.3-2 and 8.3-3.

In this section, we extend these definitions to transfer across an interface, from one well-mixed bulk phase into another different one. This case occurs more frequently than does transfer from an interface into one bulk phase; indeed, I had trouble dreaming up the examples earlier in this chapter. Transfer across an interface again sparks potentially major problems of unit conversion, but these problems are often simplified in special cases.

8.5.1 The Basic Flux Equation

Presumably, we can describe mass transfer across an interface in terms of the same type of flux equation as before:

$$N_1 = K \Delta c_1 \quad (8.5-1)$$

where N_1 is the solute flux relative to the interface, K is called an "overall mass transfer coefficient," and Δc_1 is some appropriate concentration difference. But what is Δc_1 ?

Choosing an appropriate value of Δc_1 turns out to be difficult. To illustrate this, consider the three examples shown in Fig. 8.5-1. In the first example in Figure 8.5-1(a), hot benzene is placed on top of cold water; the benzene cools and the water warms until they reach the same temperature. Equal temperature is the criterion for equilibrium, and the amount of energy transferred per time turns out to be proportional to the temperature difference between the liquids. Everything seems secure.

As a second example, shown in Fig. 8.5-1(b), imagine that a benzene solution of bromine is placed on top of water containing the same concentration of bromine. After a while, we find that the initially equal concentrations have changed, that the bromine

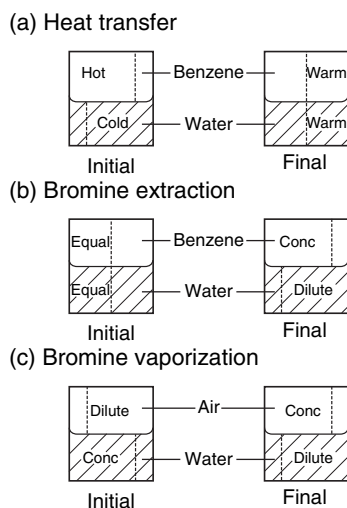


Fig. 8.5-1. Driving forces across interfaces. In heat transfer, the amount of heat transferred depends on the temperature difference between the two liquids, as shown in (a). In mass transfer, the amount of solute that diffuses depends on the solute's "solubility" or, more exactly, on its chemical potential. Two cases are shown. In (b), bromine diffuses from water into benzene because it is much more soluble in benzene; in (c), bromine evaporates until its chemical potentials in the solutions are equal. This behavior complicates analysis of mass transfer.

concentration in the benzene is much higher than that in water. This is because the bromine is more soluble in benzene, so that its concentration in the final solution is higher.

This result suggests which concentration difference we can use in Eq. 8.5-1. We should not use the concentration in benzene minus the concentration in water; that is initially zero, and yet there is a flux. Instead, we can use the concentration actually in benzene minus the concentration that would be in benzene that was in equilibrium with the actual concentration in water. Symbolically,

$$N_1 = K[c_1(\text{in benzene}) - Hc_1(\text{in water})] \quad (8.5-2)$$

where H is a partition coefficient, the ratio at equilibrium of the concentration in benzene to that in water. Note that this does predict a zero flux at equilibrium.

A better understanding of this phenomenon may come from the third example, shown in Fig. 8.5-1(c). Here, bromine is vaporized from water into air. Initially, the bromine's concentration in water is higher than that in air; afterward, it is lower. Of course, this reversal of the concentration in the liquid might be expressed in moles per liter and that in gas as a partial pressure in atmospheres, so it is not surprising that strange things happen.

As you think about this more carefully, you will realize that the units of pressure or concentration cloud a deeper truth: Mass transfer should be described in terms of the more fundamental chemical potentials. If this were done, the peculiar concentration differences would disappear. However, chemical potentials turn out to be difficult to use in practice, and so the concentration differences for mass transfer across interfaces will remain complicated by units.

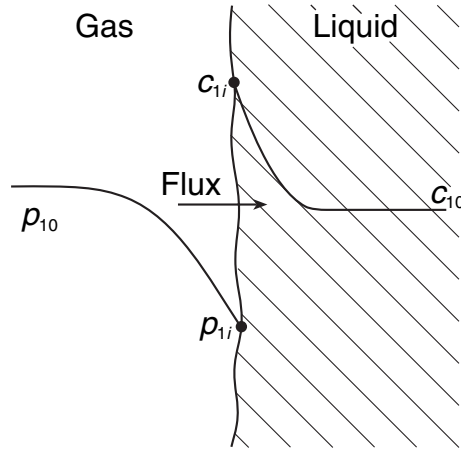


Fig. 8.5-2. Mass transfer across a gas–liquid interface. In this example, a solute vapor is diffusing from the gas on the left into the liquid on the right. Because the solute concentration changes both in the gas and in the liquid, the solute’s flux must depend on a mass transfer coefficient in each phase. These coefficients are combined into an overall flux equation in the text.

8.5.2 The Overall Mass Transfer Coefficient

We want to extend these qualitative observations in more exact equations. To do this, we consider the example of the gas–liquid interface in Fig. 8.5-2. In this case, gas on the left is being transferred into the liquid on the right. The flux in the gas is

$$N_1 = k_p(p_1 - p_{1i}) \quad (8.5-3)$$

where k_p is the gas-phase mass transfer coefficient (in, for example, $\text{mol}/\text{cm}^2 \text{ sec Pa}$), p_1 is the bulk pressure, and p_{1i} is the interfacial pressure. Because the interfacial region is thin, the flux across it will be in steady state, and the flux in the gas will equal that in the liquid. Thus,

$$N_1 = k_x(x_{1i} - x_1) \quad (8.5-4)$$

where the liquid-phase mass transfer coefficient k_x is, for example, in $\text{mol}/\text{cm}^2 \text{ sec}$, and x_{1i} and x_1 are the interfacial and bulk mole fractions, respectively.

We now need to eliminate the unknown interfacial concentrations from these equations. In almost all cases, equilibrium exists across the interface:

$$p_{1i} = Hx_{1i} \quad (8.5-5)$$

where H is a type of Henry’s law or partition constant (here in units of pressure). Combining Eqs. 8.5-3 through 8.5-5, we can find the interfacial concentrations

$$x_{1i} = \frac{p_{1i}}{H} = \frac{k_p p_1 + k_x x_1}{k_p H + k_x} \quad (8.5-6)$$

and the flux

$$N_1 = \frac{1}{1/k_p + H/k_x} (p_1 - Hx_1) \quad (8.5-7)$$

You should check the derivations of these results because they are important.

Before proceeding further, we make a quick analogy. This result is often compared to an electric circuit containing two resistances in series. The flux corresponds to the current, and the concentration difference $p_1 - Hx_1$ corresponds to the voltage. The resistance is then $1/k_p + H/k_x$ which is roughly a sum of two resistances in series. This is a good way of thinking about these effects. You must remember, however, that the resistances $1/k_p$ and $1/k_x$ are not directly added, but always weighted by partition coefficients like H .

We now want to write Eq. 8.5-7 in the form of Eq. 8.5-1. We can do this in two ways. First, we can write

$$N_1 = K_x (x_1^* - x_1) \quad (8.5-8)$$

where

$$K_x = \frac{1}{1/k_x + 1/k_p H} \quad (8.5-9)$$

and

$$x_1^* = \frac{p_1}{H} \quad (8.5-10)$$

K_x is called an “overall liquid-side mass transfer coefficient,” and x_1^* is the hypothetical liquid mole fraction that would be in equilibrium with the bulk gas. Alternatively,

$$N_1 = K_p (p_1 - p_1^*) \quad (8.5-11)$$

where

$$K_p = \frac{1}{1/k_p + H/k_x} \quad (8.5-12)$$

and

$$p_1^* = Hc_{10} \quad (8.5-13)$$

K_p is an “overall gas-side mass transfer coefficient,” and p_1^* is the hypothetical partial pressure that would be in equilibrium with the bulk liquid.

8.5.3 Details of the Partition Coefficient

The analysis above is standard, a fixture of textbooks on transport phenomena and unit operations. It can be reproduced by almost everyone who studies the subject.

However, I know that it is unevenly understood in physical terms, both by students and by experienced professionals. Indeed, explaining this repeatedly is a large part of my consulting practice. As a result, I want to pause in the standard development and go over the two tricky points again, in more detail.

The first of the tricky points comes from the basic flux equation, given by (for example)

$$N_1 = K_L (c_1^* - c_1) \quad (8.5-14)$$

In this equation, the flux across the interface is proportional to a concentration difference. The concentration difference is confusing. Clearly, c_1 is the bulk concentration of species “1,” for example, in the liquid phase. The other concentration c_1^* is harder to understand. Saying that it is “the concentration that would exist in the liquid if the other phase were in equilibrium” is true but may not help much. Alternatively, we can ask what c_1^* will be when the flux is zero. Then, c_1^* equals c_1 and we are in equilibrium.

The second tricky point in the analysis comes from the partition coefficient H . I have casually described H as a partition coefficient or a Henry’s law coefficient, neglecting the complex units which can be involved. To illustrate these, we can compare three common forms of partition coefficients, defined at equilibrium

$$p_1 = Hx_1 \quad (8.5-15)$$

$$y_1 = mx_1 \quad (8.5-16)$$

$$c_{1G} = H' c_{1L} \quad (8.5-17)$$

Here, p_1 is the partial pressure of species “1” in the gas; x_1 and y_1 are the corresponding mole fractions in the liquid and gas, respectively; and c_{1L} and c_{1G} are the molar concentrations of species “1” in the liquid and gas, respectively. The quantities H , m , and H' are all partition coefficients, and all can describe the same equilibrium between gas and liquid. However, they don’t always have the same dimensions; and they are not numerically equal even when they do have the same dimensions.

To illustrate these differences, we consider the system of oxygen gas which is partly dissolved in water. From equilibrium experiments, we find

$$p_1 = [43000 \text{ atm}] x_1 \quad (8.5-18)$$

where p_1 is the partial pressure of oxygen, and x_1 is the mole fraction of oxygen dissolved in water. To convert this into a ratio of mole fractions, we remember that

$$\frac{p_1}{p} = y_1 = \left(\frac{H}{p} \right) x_1 \quad (8.5-19)$$

Thus at atmospheric pressure, m is 42000. We can also reform Henry’s law in terms of molar concentrations

$$c_{1G} = \frac{p_1}{RT} = \left(\frac{H}{c_L RT} \right) c_{1L} \quad (8.5-20)$$

where c_L is the total molar concentration in the liquid, about 55 M for water. Because H' equals the quantity in parentheses, at 298 K, H' is 31.

We now turn to a variety of examples illustrating mass transfer across an interface. These examples have the annoying characteristic that they are initially difficult to do, but they are trivial after you understand them. Remember that most of the difficulty comes from that ancient but common curse: unit conversion.

Example 8.5-1: Oxygen mass transfer Use Equation 8.5-9 to estimate the overall liquid-side mass transfer coefficient K_x at 25 °C for oxygen from water into air. In this estimate, assume that the film thickness is 10^{-2} cm in liquids but 10^{-1} cm in gases.

Solution For oxygen in air, the diffusion coefficient is $0.23 \text{ cm}^2/\text{sec}$; for oxygen in water, the diffusion coefficient is $2.1 \cdot 10^{-5} \text{ cm}^2/\text{sec}$. The Henry's law constant in this case is $4.2 \cdot 10^4$ atmospheres. We need only calculate k_x and k_p and plug these values into Eq. 8.5-9. Finding k_x is easy:

$$\begin{aligned} k_x &= k_L c_L = \frac{D_L c_L}{0.01 \text{ cm}} = \frac{2.1 \cdot 10^{-5} \text{ cm}^2/\text{sec}}{0.01 \text{ cm}} \left(\frac{\text{mol}}{18 \text{ cm}^3} \right) \\ &= 1.2 \cdot 10^{-4} \text{ mol/cm}^2 \text{ sec} \end{aligned}$$

Finding k_p involves the unit conversions given in Table 8.2-2:

$$\begin{aligned} k_p &= \frac{k_G}{RT} = \frac{D_G}{(0.1 \text{ cm})(RT)} \\ &= \frac{0.23 \text{ cm}^2/\text{sec}}{(0.1 \text{ cm}) \left(82 \text{ cm}^3 \text{ atm/mol K} \right) (298 \text{ K})} \\ &= 9.4 \cdot 10^{-5} \text{ mol/cm}^2 \text{ sec atm} \end{aligned}$$

Inserting these results into Eq. 8.5-9, we find

$$\begin{aligned} K_x &= \frac{1}{1/k_x + 1/k_p H} \\ &= \frac{1}{\left(\frac{\text{cm}^2 \text{ sec}}{1.2 \cdot 10^{-4} \text{ mol}} \right) + \frac{\text{cm}^2 \text{ sec atm}}{9.4 \cdot 10^{-5} \text{ mol} (43000 \text{ atm})}} \\ &= 1.2 \cdot 10^{-4} \text{ mol/cm}^2 \text{ sec} \end{aligned}$$

The mass transfer is completely dominated by the liquid-side resistance. This would also be true if we calculated the overall gas-side mass transfer coefficient, K_p . Gas absorption is commonly controlled by mass transfer in the liquid and is one reason that reactive liquids are effective.

Example 8.5-2: Benzene mass transfer Estimate the overall liquid-side mass transfer coefficient in the distillation of benzene and toluene. At the concentrations used, you expect a temperature of 90 °C and (at equilibrium)

$$y_1^* = 0.70x_1 + 0.39$$

The molar volume of the liquid is about 97 cm³/mol. As in the previous problem, assume that the film thickness in the liquid is 0.01 cm and in the gas is 0.1 cm.

Solution In the vapor, the diffusion coefficient is about 0.090 cm²/sec; in the liquid, it is 1.9 · 10⁻⁵ cm²/sec. As in the previous example,

$$\begin{aligned} k_x &= k_L c_L = \frac{D_L c_L}{0.01 \text{ cm}} \\ &= \frac{1.9 \cdot 10^{-5} \text{ cm}^2/\text{sec} \left(\frac{1 \text{ mol}}{97 \text{ cm}^3} \right)}{0.01 \text{ cm}} \\ &= 2.0 \cdot 10^{-5} \frac{\text{mol}}{\text{cm}^2 \text{ sec}} \end{aligned}$$

If the distillation is at atmospheric pressure, the Henry's law constant is 0.70 atm (cf. Equations 8.5-18 and 8.5-19). Then

$$\begin{aligned} k_p &= \frac{k_G}{RT} = \frac{D_G}{0.1 \text{ cm} (RT)} \\ &= \frac{0.090 \text{ cm}^2/\text{sec}}{0.1 \text{ cm} \left(82 \text{ cm}^3 \text{ atm/mol K} \right) (363 \text{ K})} \\ &= 3.0 \cdot 10^{-5} \text{ mol/cm}^2 \text{ sec atm} \end{aligned}$$

Inserting these results into Equation 8.5-9 gives

$$\begin{aligned} K_x &= \frac{1}{1/k_x + 1/k_p H} \\ &= \frac{1}{\left(\frac{\text{cm}^2 \text{ sec}}{2.0 \cdot 10^{-5} \text{ mol}} \right) + \frac{\text{cm}^2 \text{ sec atm}}{3.0 \cdot 10^{-4} \text{ mol} (0.70 \text{ atm})}} \\ &= 1.8 \cdot 10^{-5} \text{ mol/cm}^2 \text{ sec} \end{aligned}$$

The distillation is about equally controlled by the liquid and the vapor. This is typical.

Example 8.5-3: Perfume extraction Jasnone (C₁₁H₁₆O) is a valuable material in the perfume industry, used in many soaps and cosmetics. Suppose we are recovering this

material from a water suspension of jasmine flowers by an extraction with toluene. The aqueous phase is continuous, with suspended flowers and toluene droplets. The mass transfer coefficient in the toluene droplets is $3.0 \cdot 10^{-4} \text{ cm}^2/\text{sec}$; the mass transfer coefficient in the aqueous phase is $2.4 \cdot 10^{-3} \text{ cm}^2/\text{sec}$. Jasmone is about 150 times more soluble in toluene than in the suspension. What is the overall mass transfer coefficient?

Solution For convenience, we designate all concentrations in the toluene phase with a prime and all those in the water without a prime. The flux is

$$N_1 = k(c_{10} - c_{1i}) = k'(c'_{1i} - c'_{10})$$

The interfacial concentrations are in equilibrium:

$$c'_{1i} = Hc_{1i}$$

Eliminating these interfacial concentrations, we find

$$N_1 = \left[\frac{1}{1/k' + H/k} \right] (Hc_{10} - c'_{10})$$

The quantity in square brackets is the overall coefficient K' that we seek. This coefficient is based on a driving force in toluene. Inserting the values,

$$\begin{aligned} K' &= \frac{1}{\frac{1}{3.0 \cdot 10^{-4} \text{ cm}^2/\text{sec}} + \frac{150}{2.4 \cdot 10^{-3} \text{ cm}^2/\text{sec}}} \\ &= 1.5 \cdot 10^{-5} \text{ cm}^2/\text{sec} \end{aligned}$$

Similar results for the overall coefficient based on a driving force in water are easily found.

Two points about this problem deserve mention. First, the result is a complete parallel to Eq. 8.5-12, but for a liquid–liquid interface instead of a gas–liquid interface. Second, mass transfer in the water dominates the process even though the mass transfer coefficient in water is larger because jasmone is so much more soluble in toluene.

Example 8.5-4: Overall mass transfer coefficients in a packed tower We are studying gas absorption into water at 2.2 atm total pressure in a packed tower containing Berl saddles. From earlier experiments with ammonia and methane, we believe that for both gases the mass transfer coefficient times the packing area per tower volume is 18 lb mol/hr ft³ for the gas side and 530 lb mol/hr ft³ for the liquid side. The values for these two gases may be similar because methane and ammonia have similar molecular weights. However, their Henry's law constants are different: 75 atm for ammonia and 41,000 atm for methane. What is the overall gas-side mass transfer coefficient for each gas?

Solution This is essentially a problem in unit conversion. Although you can extract the appropriate equations from the text, I always feel more confident if I repeat parts of the derivation.

The quantity we seek, the overall gas-side transfer coefficient K_y , is defined by

$$\begin{aligned} N_1 a &= K_y a (y_1 - y_1^*) \\ &= k_y a (y_1 - y_{1i}) \\ &= k_x a (x_{1i} - x_1) \end{aligned}$$

where y_1 and x_1 are the gas and liquid mole fractions.

The interfacial concentrations are related by Henry's law:

$$p_{1i} = py_{1i} = Hx_{1i}$$

When these interfacial concentrations are eliminated, we find that

$$\frac{1}{K_y a} = \frac{1}{k_y a} + \frac{H/p}{k_x a}$$

In passing, we recognize that y_1^* must equal Hx_{10}/p .

We can now find the overall coefficient for each gas. For ammonia,

$$\frac{1}{K_y a} = \frac{1}{18 \text{ lb mol/hr ft}^3} + \frac{75 \text{ atm}/2.2 \text{ atm}}{530 \text{ lb mol/hr ft}^3}$$

$$K_y a = 8.3 \text{ lb mol/hr ft}^3$$

The overall resistance is affected by resistances in both the gas and the liquid. For methane,

$$\frac{1}{K_y a} = \frac{1}{18 \text{ lb mol/hr ft}^3} + \frac{41,000 \text{ atm}/2.2 \text{ atm}}{530 \text{ lb mol/hr ft}^3}$$

$$K_y a = 0.03 \text{ lb mol/hr ft}^3$$

The overall mass transfer coefficient for methane is smaller and is dominated by the liquid-side mass transfer coefficient.

8.6 Conclusions

This chapter presents an alternative model for diffusion, one using mass transfer coefficients rather than diffusion coefficients. The model is most useful for mass transfer across phase boundaries. It assumes that large changes in the concentration occur only very near these boundaries and that the solutions far from the boundaries are well mixed. Such a description is called a lumped-parameter model.

Mass transfer coefficients provide especially useful descriptions of diffusion in complex multiphase systems. They are basic to the analysis and design of industrial processes like absorption, extraction, and distillation. Mass transfer coefficients are not useful in chemistry when the focus is on chemical kinetics or chemical change. They are not useful in studies of the solid state, where concentrations vary with both position and time, and lumped-parameter models do not help much.

In this chapter, we have shown how experimental results can be analyzed in terms of mass transfer coefficients. We have also shown how values of these coefficients can be efficiently organized as dimensionless correlations, and we have cataloged published correlations that are commonly useful. These correlations are complicated by problems with units that come out of a plethora of closely related definitions. These complications

are most confusing for mass transfer across fluid–fluid interfaces, from one well-mixed phase into another.

All in all, the material in this chapter is a solid alternative for analyzing diffusion near interfaces. It is basic stuff for chemical engineers, but it is an unexplored method for many others. It repays careful study.

Questions for Discussion

1. What are the dimensions of a mass transfer coefficient?
2. What are the dimensions of a diffusion coefficient?
3. How can you convert from a mass transfer coefficient based on a concentration difference to one based on a partial pressure difference?
4. Which is typically bigger, a mass transfer coefficient in a liquid or one in a gas?
5. Why do overall mass transfer coefficients vary with partition coefficients, but overall heat transfer coefficients do not?
6. If the flow doubles, how much will the mass transfer coefficient typically change?
7. If the diffusion coefficient doubles, how much does the mass transfer coefficient change?
8. If you have a system with the same solute concentration everywhere, will mass transfer occur?
9. When will a bulk concentration equal an interfacial concentration? When won't it?
10. What are typical units for partition coefficients?

Problems

1. A wet t-shirt hung on a hanger has a total surface area of about 0.6 m². It loses water as follows:

time (pm)	Weight (g)
3:15	661
3:20	640
3:48	580
4:00	553

If the saturation vapor pressure of the water of 20 mm Hg gives a water concentration of 20 g/m³ and the room has a relative humidity of 30%, estimate the mass transfer coefficient from the t-shirt.

2. Water flows through a thin tube, the walls of which are lightly coated with benzoic acid. The benzoic acid is dissolved very rapidly and so is saturated at the pipe's wall. The water flows slowly, at room temperature and 0.1 cm/sec. The pipe is 1 cm in diameter. Under these conditions, the mass transfer coefficient k varies along the pipe:

$$\frac{kd}{D} = 1.62 \left(\frac{d^2 v}{DL} \right)^{1/3}$$

where d and L are the diameter and length of the pipe and v is the average velocity in the pipe. What is the average concentration of benzoic acid in the water after 2 m of pipe?

3. Water containing 0.1-M benzoic acid flows at 0.1 cm/sec through a 1-cm-diameter rigid tube of cellulose acetate, the walls of which are permeable to small electrolytes. These walls are 0.01 cm thick; solutes within the walls diffuse as through water. The tube is immersed in a large well-stirred water bath. Under these conditions, the flux of benzoic acid from the bulk to the walls can be described by the correlation in Problem 8.2. After 50 cm of tube, what fraction of a 0.1-M benzoic acid solution has been removed? Remember that there is more than one resistance to mass transfer in this system.
4. How much is the previous answer changed if the benzoic acid solution in the tube is in benzene, not water?
5. A disk of radioactively tagged benzoic acid 1 cm in diameter is spinning at 20 rpm in 94 cm³ of initially pure water. We find that the solution contains benzoic acid at $7.3 \cdot 10^{-4}$ g/cm³ after 10 hr 4 min and $3.43 \cdot 10^{-3}$ g/cm³ after a long time (i.e., at saturation). (a) What is the mass transfer coefficient? *Answer:* $8 \cdot 10^{-4}$ cm/sec. (b) How long will it take to reach 14% saturation? (c) How closely does this mass transfer coefficient agree with that expected from the theory in Example 3.4-3?
6. As part of the manufacture of microelectronic circuits, silicon wafers are partially coated with a 5,400-Å film of a polymerized organic film called a photoresist. The density of this polymer is 0.96 g/cm³. After the wafers are etched, this photoresist must be removed. To do so, the wafers are placed in groups of twenty in an inert "boat," which in turn is immersed in strong organic solvent. The solubility of the photoresist in the solvent is $2.23 \cdot 10^{-3}$ g/cm³. If the photoresist dissolves in 10 minutes, what is its mass transfer coefficient? (S. Balloge) *Answer:* $4 \cdot 10^{-5}$ cm/sec.
7. You are studying mass transfer of a solute from a gas across a gas-liquid interface into a reactive liquid. The mole fraction in the bulk gas is 0.01; that in the bulk liquid is 0.00; and the equilibrium across the interface is

$$y_i^* = \frac{x_i}{3}$$

The individual mass transfer coefficient k_y and k_x are 0.50 and 0.60 mol/m² sec, respectively. (a) What is the interfacial concentration in the vapor? (b) Sketch, to scale, the mole fractions in both phases across the interface.

8. Calculate the fraction of the resistance to SO₂ transport in the gas and liquid membrane phases for an SO₂ scrubber operating at 100 °C. The membrane liquid, largely ethylene glycol, is $5 \cdot 10^{-3}$ cm thick. In it, the SO₂ has a diffusion coefficient of about $0.85 \cdot 10^{-5}$ cm²/sec and a solubility of

$$\frac{0.026 \text{ mol SO}_2/\text{l}}{\text{mm Hg of SO}_2}$$

In the stack gas, there is an unstirred film adjacent to the membrane 0.01 cm thick, and the SO₂ has a diffusion coefficient of 0.13 cm²/sec. (W. J. Ward)

9. Estimate the average mass transfer coefficient for water evaporating from a film falling at 0.82 cm/sec into air. The air is at 25 °C and 2 atm, and the film is

186 cm long. Express your result in cubic centimeters of H_2O vapor at STP/(hr cm^2 atm).

10. Find the dissolution rate of a cholesterol gallstone 1 cm in diameter immersed in a solution of bile salts. The solubility of cholesterol in this solution is about $3.5 \cdot 10^{-3} \text{ g/cm}^3$. The density difference between the bile saturated with cholesterol and that containing no cholesterol is about $3 \cdot 10^{-3} \text{ g/cm}^3$; the kinematic viscosity of this solution is about $0.06 \text{ cm}^2/\text{sec}$; the diffusion coefficient of cholesterol is $1.8 \cdot 10^{-6} \text{ cm}^2/\text{sec}$. *Answer: 0.2 g per month.*
11. Air at 100°C and 2 atm is passed through a bed 1 cm in diameter composed of iodine spheres 0.07 cm in diameter. The air flows at a rate of 2 cm/sec, based on the empty cross-section of bed. The area per volume of the spheres is $80 \text{ cm}^2/\text{cm}^3$, and the vapor pressure of the iodine is 45 mm Hg. How much iodine will evaporate from a bed 13 cm long, assuming a bed porosity of 40%? *Answer: 5.6 g/hr.*
12. The largest liquid–liquid extraction process is probably the dewaxing of lubricants. After they are separated by distillation, crude lubricant stocks still contain significant quantities of wax. In the past, these waxes were precipitated by cooling and separated by filtration; now, they are extracted with mixed organic solvents. For example, one such process uses a mixture of propane and cresylic acid. You are evaluating a new mixed solvent for dewaxing that has physical processes like those of catechol. You are using a model lubricant with properties characteristic of hydrocarbons. Waxes are 26.3 times more soluble in the extracting solvent than they are in the lubricant. You know from pilot-plant studies that the mass transfer coefficient based on a lubricant-side driving force is

$$K_L a = 16,200 \text{ lb/ft}^3 \text{ hr}$$

What will it be (per second) if the driving force is changed to that on the solvent side?

13. You need to estimate an overall mass transfer coefficient for solute adsorption from an aqueous solution of density 1.3 g/cm^3 into hydrogel beads 0.03 cm in diameter. The coefficient sought K_y is defined by

$$N_1 = K_y(y - y^*)$$

where N_1 has the units of $\text{g/cm}^2 \text{ sec}$, and the y 's have units of solute mass fraction in the water. The mass transfer coefficient k_S in the solution is 10^{-3} cm/sec ; that within the beads is given by

$$k_B = \frac{6D}{d}$$

where d is the particle diameter and D is the diffusion coefficient, equal here to $3 \cdot 10^{-6} \text{ cm}^2/\text{sec}$. Because the beads are of hydrogel, the partition coefficient is one. Estimate K_y in the units given.

14. A horizontal pipe 10 inches in diameter is covered with an inch of insulation that is 36% voids. The insulation has been soaked with water. The pipe is now drying slowly and hence almost isothermally in 80°F air that has a relative humidity of about 55%. Estimate how long it will take the pipe to dry, assuming that capillarity always brings any liquid water to the pipe's surface. (H. A. Beesley)

Further Reading

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- Seader, J. D. and Henley, E. J. (2006). *Separation Process Principles*, 2nd ed. New York: Wiley.